

Fatemeh (Naz) Hadavandkhani

HCI Researcher · UI/UX Designer · Creative Technologist

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Boulder, Colorado

Research Interests

Human–Computer Interaction (HCI), Human–AI Interaction, Mixed and Augmented Reality, Generative AI, Proactive AI Systems, UI/UX Research and Design, Interactive Systems, and Research through Design.

Education

University of Colorado Boulder, ATLAS Institute

Boulder, CO

M.S. in Creative Technology & Design

2024–2026

M.S. Thesis: *Proactive AR/AI for Everyday Activities*

Context-aware AI assistance for everyday activities in augmented and mixed reality. Manuscript in preparation; planned submission November 2026.

Art University of Tehran

Tehran, Iran

M.S. in Art Research

2021–2023

Thesis: *Study of Women’s Role in Protest Movements through Visual Arts in Iran during the Qajar and Pahlavi Dynasties*

University of Tehran

Tehran, Iran

B.Sc. in Food Machinery (Mechanical) Engineering

2014–2019

Research & Manuscripts

HyperVision: Designing In-Situ Interaction for Generative Re-Rendering in Mixed Reality

Suibi Che-Chuan Weng, Anika Mahajan, **Fatemeh Hadavandkhani**, Pritalee Kadam, Yan-Ming Chiou, Amy Banić, Ryo Suzuki, Ellen Yi-Luen Do.

Submitted to ACM IUI 2027.

Developed and evaluated an intelligent mixed-reality system that enables generative transformation of users’ passthrough environments through text, sketches, masks, and virtual elements. A within-subjects study with 20 participants examined the effects of transformation strength on recognizability, creativity, coherence, and perceived control.

DreamPortal: Generative In-Situ Behavior Authoring in Cyber-Physical Environments through Mixed Reality

Suibi Che-Chuan Weng, **Fatemeh Hadavandkhani**, Ryo Suzuki, Ellen Yi-Luen Do.

Submitted to ACM CHI 2027.

Investigated natural-language authoring of behaviors involving virtual objects, physical environments, and IoT devices in mixed reality. The project used a Research through Design approach, iterative prototyping, and a 20-participant user study to examine how users express, execute, inspect, revise, and compose interactive behaviors.

Current Research

Proactive AR/AI for Everyday Activities

M.S. Thesis, 2026

- Investigating proactive, context-aware AI assistance in augmented and mixed reality for everyday activities.
- Exploring how multimodal systems can recognize relevant changes in a user’s environment and provide assistance when the user’s attention is directed elsewhere.
- Developing the project with faculty and PhD collaborators at CU Boulder; manuscript planned for submission in November 2026.

Selected UI/UX & Interactive Projects

RayBrella — UI/UX, Product Design, Physical Computing

Designed a portable UV-protective parasol integrating solar charging, physical interaction, and a digital user interface. [Figma Prototype]

Memento — UI/UX Research and Mobile Application Design

Conducted user research and designed a mobile interface supporting the planning of leisure activities and shared experiences.

STEMblazers — UX Research and UI Design

Conducted user research and developed interface recommendations for the STEMblazers website.

Spotify — UI/UX Research

Studied listening flows, personalization, and social interaction to identify opportunities for improving the user experience.

Haptic Memory Game — Interaction Design and Haptics

Designed a multisensory memory game exploring tactile interaction and learning.

Robotic Lawn Mower — Engineering and Prototyping

Designed and prototyped a tracked autonomous lawn mower with emphasis on safety and mechanical durability.

Experience

University of Colorado Boulder

Boulder, CO

Digital Fabrication Lab Assistant

Oct. 2024–Present

- Support students in prototyping, fabrication, and iterative development of physical and interactive projects.
- Operate and assist with digital fabrication technologies including laser cutters, FDM and resin 3D printers, vinyl cutters, and related prototyping equipment.
- Provide technical guidance on preparing digital designs for fabrication and troubleshooting fabrication workflows.

Technical & Design Skills

Programming	Python, Java, JavaScript / p5.js
UI/UX & Research	Figma, Miro, user research, usability testing, prototyping, interaction design
XR & Interactive Systems	Mixed Reality, Augmented Reality, Arduino, physical computing
3D Design & Fabrication	Fusion 360, Rhino, 3D printing, resin printing, laser cutting, vinyl cutting
Visual Design	Adobe Illustrator

Selected Interactive Experiments

Chicken Flock | Fox Logistics | Escape from the Zoo

Small interactive game prototypes exploring game mechanics, interaction design, and implementation.

Honors

- Ranked **1st** in the M.S. in Art Research program, Art University of Tehran, 2023.
- Ranked in the **top 2%** of Iran's National University Entrance Examination for undergraduate studies, 2014.

Additional Experience

Radiacar Co.

Tehran, Iran

Digital Marketing Associate

2021–2023

Product photography, digital content, social media management, and customer-facing digital communication.